DOMINIC IRLA

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EDUCATION

University of Illinois at Chicago - Bachelor of Science in Computer Science *Expected May 2027*

- 3.84 GPA. Coursework includes UX Design, Software Design, webGPU, and Algorithmic Practices. Dean's List.

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, HTML/CSS, SQL, Python, F#, GO, Java, C, C++, SQL
Frameworks & Libraries: React, Tailwind, Vite, Astro, Three.js, WebGPU, Canvas 2D API, Web Bluetooth. Laravel, NodeJS
Tools: Git, REST APIs, Firebase, PostgreSQL, Cursor, Claude, Figma, Supabase, Shibboleth

WORK EXPERIENCE

Bytedance/Tiktok *May 2026 -- August 2026*
Frontend Software Engineer Intern

- Incoming summer intern on the Global Payment team, building frontend systems in **React/TypeScript** for a high-scale international payment platform.
- Will implement responsive, accessible UI components and collaborate with engineers to integrate payment API into the user product.

UIC College of Engineering *June 2024 -- Present*
Web Developer

- Build and ship full-stack web applications using **React, Laravel, and MySQL** supporting day-to-day faculty and staff operations.
- Replaced a disorganized email workflow with a modern travel request system in **React**, improving approval tracking and cutting turnaround time.
- Automated CSV-to-financial-report pipeline in **TypeScript**, eliminating **3–4 weeks** of manual work per quarter for staff.
- Own ongoing maintenance and iteration based on user feedback — catching bugs, shipping improvements, and keeping applications reliable in production.

EXTRACURRICULAR ACTIVITY

UIC Research - Wearable Technology & Sensory Environment Lab *March 2026 -- Present*
Web Developer

- Took ownership of a serious-games research project and rebuilt the frontend using **React TypeScript (Vite)**, enabling patients to interact with therapeutic games via a custom tongue trackpad. Used canvases to display games in real time.
- Integrated the tongue trackpad device over **Web Bluetooth (BLE/GATT)** — streaming live sensor input from hardware directly into the browser with no backend, handling device pairing, characteristic subscriptions, and real-time data parsing.
- Collaborated with Biomedical Engineering majors and designed around strict data constraints (HIPAA) — no live service or database; all state handled entirely client-side.

Sparkhacks *July 2025 -- Present*
Web Developer

- Building the application portal for Chicago's largest hackathon **(400+ entries)** using **Astro and Firebase**.
- Developed full-stack features for registration, scheduling, and submissions — used by attendees, sponsors, and judges.
- Collaborate with 20 student organizers to deliver a responsive, accessible experience under a real deadline.

Association for Computer Machinery - UIC Chapter *Sept 2025 -- Present*
SIG Lead – Web Development

- Lead a team building LinkIt – a live, deployed web app where users create/join rooms to collaboratively organize links, built with React and Supabase (real-time DB + auth).
- Run workshops on frontend frameworks, REST APIs, ORMs, and databases; mentor members on code quality and system design.

PROJECTS

Neural Field Simulator | *JavaScript, WebGPU, React* *April 2026*

- Built a real-time browser-based simulation of neural activity using **WebGPU**, rendering how neurons respond to external stimuli.
- Implemented GPU compute shaders for parallel simulation, achieving smooth real-time frame rates in the browser.
- Designed an interactive UI in **React** allowing users to adjust stimulus parameters and observe dynamic field responses live.

Automated Drone Vision | *Python, Embedded Systems, YOLO* *August 2025*

- Wrote **Python** for an **Nvidia Jetson**-controlled drone to autonomously identify and retrieve targets using **YOLO** object detection.
- Designed the EDT Air Team's public website with real-time updates and 3D component renders in **Three.js**.

AWARDS / CERTIFICATES

1st Place | UX Design Contest | *FBLA, IL* *July 2023*

- Earned 1st place at the state level, demonstrating applied knowledge of UX design principles, user research, and interface design.